

DOMINICK BRAICO

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Education

University of Michigan Ann Arbor

Expected December 2027

M.S. Robotics

University of Illinois Urbana-Champaign

May 2026

B.S. Mechanical Engineering, Minor in Computer Science

GPA: 3.91

Relevant Coursework: Mobile Manipulation, Software Development for Mobile Robotics, Robot Kinematics and Dynamics, Robotic Systems Laboratory, Mathematics for Robotics, Signals and Systems

Professional Experience

Graduate Robotics Research Assistant

May 2026 – Aug 2026

Carnegie Mellon University Robotics Institute - Matt Lab

- Designed the entire mechanical system for an autonomous server-disassembly robotic workcell — slated for deployment at Apple's Austin recycling facility
- Built a computer-vision pipeline (YOLO) that localizes server components and streams object coordinates to the manipulation stack for removal and storage
- Communicated engineering progress through bi-weekly presentations, bridging the gap between technical data and stakeholder understanding

Controls Engineering Intern

May 2025 – Aug 2025

Whisper Aero

- Developed real-time firmware in C/C++ implementing field-oriented control for high-power-density 3-phase motors, designing a live Python GUI for auto-tuning and live performance tracking
- Automated the dynamometer test sequence in Python — data acquisition, logging, and power/thermal characterization across the propulsor duty cycle — replacing manual runs with a repeatable pipeline

Robotics R&D Intern

May 2024 – March 2025

Construction Engineering Research Laboratory - Robotics for Engineer Operations

- Led systems-level integration and validation of the Co3MaNDR cable-driven robot, debugging hardware/software interfaces across mechanical, electrical, and autonomy subsystems
- Designed a ruggedized electronic control module housing robot sensors and the compute unit, reducing electrical-system volume and improving weatherproofing
- Modeled and fabricated sensor mounts in SolidWorks, improving sensor-data integrity on heavy-equipment mobile robots

Projects

Autonomous Surface Vessel (ASNE PEP26)

- Built an autonomous navigation stack on ArduPilot with GPS/IMU sensor fusion for waypoint guidance; placed 5th of 17 collegiate teams
- Developed an OpenCV vision pipeline for buoy detection and boat localization, feeding course-navigation and collision-avoidance logic

Robotic Mobile Manipulator (Software Development for Mobile Robotics Coursework)

- Built a Gazebo simulation of a 6 DOF industrial robotic arm mounted to a Clearpath Husky mobile platform
- Integrated simulated wheel encoders and lidar, developing a ROS 2 node network for state estimation
- Implemented custom ROS 2 controllers for joint-level actuator commands

Electric Motorcycle Control Systems & Sensor Integration (GHOST Electric Motorcycles)

- Cofounded an electric motorcycle design team, leading sensor integration and motor controls for the club's first design
- Configured a CAN bus network for communication between motor controller, BMS, and onboard sensors; performed bench testing to validate electrical system performance prior to vehicle integration

Skills

Software: Python, C/C++, ROS 2, Git, Linux, OpenCV, ArduPilot, Gazebo, Isaac Sim

Mechanical Design: SolidWorks (Certified Mechanical Design Associate), Autodesk Fusion, GD&T (ASME Y14.5)

Fabrication: 3D Printing, Waterjet, Manual Lathe & Mill, Shop Tools, Soldering